

Name: _____ Date: _____

Period: _____

Parametric Planar Motion Practice

1. Let $f(t) = (t^2 - 2t, 3t - t^2)$ for $0 \leq t \leq 4$.

(a) Complete the table.

t	$x(t) = t^2 - 2t$	$y(t) = 3t - t^2$	(x, y)
0	_____	_____	_____
1	_____	_____	_____
2	_____	_____	_____
3	_____	_____	_____
4	_____	_____	_____

(b) Find the horizontal extrema of the particle's motion. Use the vertex formula for $x(t)$ and check endpoints. Show work. _____

(c) Find the vertical extrema. Use the vertex formula for $y(t)$ and check endpoints. _____

2. Using $f(t) = (t^2 - 2t, 3t - t^2)$, $0 \leq t \leq 4$:

(a) Find all values of t where $y(t) = 0$. Then state the x -intercepts of the curve. _____

(b) Find all values of t where $x(t) = 0$. Then state the y -intercepts of the curve. _____

Contextual Problem

3. A ball is hit and moves with $x(t) = 4t$ (meters east) and $y(t) = -5t^2 + 20t$ (meters high) for $0 \leq t \leq 4$.

(a) At what time does the ball reach its maximum height? What is that height? _____

(b) The ball hits the ground when $y(t) = 0$. Find the time this occurs (other than $t = 0$) and state the ball's horizontal position at that moment. _____

AP-Style Practice

4. A particle moves with $h(t) = (t^2 - 2t - 3, t^2 - 4)$ for $-1 \leq t \leq 4$. Which of the following correctly identifies the y -intercepts of the parametric curve?

- (A) $(-4, 0)$ and $(5, 0)$
- (B) $(0, -3)$ and $(0, 5)$
- (C) $(0, -3)$ only
- (D) $(0, -4)$ and $(0, 12)$